6.5.2.2 Kaan (TF-X) – 2023

The Kaan (TF-X) represents Turkey's first indigenously developed fifth-generation multirole combat aircraft, designed and manufactured by Turkish Aerospace Industries (TAI). Officially renamed *Kaan* in 2023, the aircraft embodies Turkey's ambition to achieve strategic independence in aerospace defense while entering the league of nations capable of developing fifth-generation fighter technology. Developed under the guidance of the Turkish Presidency of Defense Industries (SSB), Kaan is envisioned as a stealthy, AI-driven, and optionally manned aircraft capable of performing both air superiority and deep-strike missions across modern battle environments (Kanat, et al. 2025).

The Kaan's design integrates a twin-engine stealth configuration, featuring a blended fuselage, internal weapon bays, and radar-absorbent coatings for reduced radar cross-section (RCS). Its aerodynamic layout, comprising diamond-shaped wings and forward-angled tail surfaces, enhances maneuverability, high-speed stability, and lift generation. The stealth optimization and advanced composite materials used in construction provide superior endurance, agility, and survivability (Figure 6.2).

